

Distinguishing “Indistinguishable” Traffic: An Analysis of the Fingerprints left by “Invisible” Browsing

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1.1 Project Summary

The Tor project is an open-source Internet browser application created with the intent to provide access to the Internet with a claimed “untraceable” privacy that the creators of the application believe is something not afforded to consumers by typical web browsing applications. “Snowflake” is an extension of the Tor application designed to allow users to access Tor on networks where it has been explicitly blocked, and hide the fact the user is using Tor at all, which is something the base application does not do: Tor hides where the user’s traffic is going to/from, so network administrators can only see that they are using Tor, but not the fact that the user is accessing any specific website or web application. In 2020, Kyle MacMillan, Jordan Holland, Prateek Mittal published a paper analyzing the “fingerprintability” of Snowflake traffic on any given network, as opposed to other applications that use WebRTC (Web Real-Time Communication) to facilitate browser-to-browser connections. This paper and its results are important twofold. On one hand, it may be deemed necessary to distinguish Snowflake/Tor traffic on any given network to identify individuals that are trying to circumvent standard network administration. On the other, the knowledge that that traffic can indeed be distinguished from similar WebRTC traffic is important to the developers of Snowflake and Tor so they can further develop systems to facilitate and protect user data privacy in a political climate that seems to be heading towards continued and advanced mass surveillance enabled by AI and ML applications.

1.2 Data

The objective of our project is to reproduce the results and understand the methods in *Evaluating snowflake as an indistinguishable censorship circumvention tool* (MacMillan, Holland, and Mittal, 2020), therefore we will use the dataset linked in the research paper’s [Google Drive](#).

1.3 Machine Learning Models

We expect to try scikit-learn’s GradientBoostingClassifier, which is an XGBoost model. Since a random forest classifier was used in the research paper, we thought that selecting a different ensemble method would be interesting in terms of how a bagging versus boosting process would impact the difficulty of tuning the model to achieve a comparable F1 score.

1.4 Evaluation

To evaluate our models, we will report accuracy, precision, recall, F1 score, AUC, and any other relevant metrics, then compare our results to the MacMillan paper. As an interesting ablation, we could also actively intervene by removing fields from the DTLS handshakes in order to establish

a causal link between certain features and model performance (which features help the model the most? Which are the least necessary?)

1.5 Learning Objectives

Our expectations are to learn not only whether or not we are able to replicate the 99.8% accuracy with which the original authors were able to distinguish Snowflake traffic from the original dataset, but familiarize ourselves with training an AI model, and the training / deployment pipeline. This seems to be a relatively complicated process, even with all the tools at our disposal, and we are excited to dive into learning about how to go about deploying different applications of Machine Learning technology.